

ACDC: Interpretability on Actual Clinical data using Domain-Specific & Causally validated Sparse Autoencoders

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Abstract

Sparse autoencoders (SAEs) are widely used to decompose neural network representations into interpretable features, but standard evaluation metrics fail to distinguish trained features from random baselines (Korznikov et al., 2026; Heap et al., 2025). These metrics operate entirely within the model’s own computational graph; what is missing is validation against external ground truth. We introduce *concordance validation*, a methodology that tests whether unsupervised LLM-generated feature explanations align with external structured labels that the explanation pipeline never observed. We train SAEs on Gemma-2-2B residual-stream activations from 50,000 MIMIC-IV discharge summaries and validate features against ICD-9 diagnosis codes through three modalities: statistical grounding (point-biserial r up to 0.864 with specific codes), semantic concordance (explanation–label alignment rises from 85% to 98.6% as correlation strength increases), and causal ablation (grounded features produce differential loss on diagnosis-positive notes; median Cliff’s $\delta=0.30$ vs. -0.04 for controls). Features above $r > 0.6$ each track exactly one diagnosis code. Domain-trained SAEs outperform a published general-domain SAE by $13\times$ in grounded feature count at $r > 0.3$, a gap we attribute to feature-direction mismatch rather than distributional shift. We release our code and validation framework.

1 Introduction

Sparse autoencoders (SAEs) have become the dominant approach to decomposing neural network representations into interpretable features. Foundational work established that SAEs can recover monosemantic directions from transformer activations, outperforming alternative decomposition methods in interpretability and enabling causal intervention (Bricken et al., 2023; Cunningham et al., 2023). Subsequent efforts have scaled SAE training to large-scale models (Gao et al., 2024; Raja-

manoharan et al., 2024; Lieberum et al., 2024) and extended the paradigm toward circuit-level attribution (Shu et al., 2025; Ameisen et al., 2025). However, a growing body of critical work has challenged whether standard evaluation metrics (auto-interpretability scores, reconstruction fidelity, sparse probing, and causal editing) can reliably distinguish genuine learned features from artifacts of the decomposition method itself (Korznikov et al., 2026; Heap et al., 2025; Leask et al., 2025; Ma et al., 2026; McCann, 2026). These critiques share a reliance on *internal* metrics: evaluation conducted within the model’s own computational graph or scored by LLMs operating on the same activations. The field would benefit from evaluation methods that validate SAE features against *external* ground truth that exists independently of both the model and the decomposition.

Clinical text provides a particularly well-suited setting for this kind of validation. Hospital discharge summaries describe diagnoses, treatments, and patient trajectories in dense specialized language, and each clinical encounter is independently annotated with structured ICD-9 diagnosis codes by human medical coders. These codes capture core clinical concepts without being derived from any automated text process (Johnson et al., 2023b,a). The MIMIC-IV dataset pairs such unstructured notes with their structured codes at scale, providing an opportunity to test whether SAE features recover meaningful domain concepts in a setting where interpretive reliability matters. If a general-purpose language model’s representations of clinical text encode clinical structure, and if an SAE recovers that structure as individual features, then those features should be verifiable against the structured labels: they should correlate with specific diagnosis codes, their unsupervised LLM-generated explanations should align with those codes, and ablating them should differentially affect the model’s predictions on relevant notes.

We propose *concordance validation*, a methodology that checks whether unsupervised feature explanations converge with external structured labels the explanation pipeline never observed. We demonstrate it by training two SAE architectures (JumpReLU and vanilla ReLU; Rajamanoharan et al., 2024; Bricken et al., 2023) on Gemma-2-2B layer-16 residual stream activations from 50,000 MIMIC-IV discharge summaries. Concordance is evaluated at multiple correlation thresholds, yielding a *concordance gradient*: as the statistical association between a feature and a diagnosis code strengthens, the rate at which the feature’s LLM explanation matches that code should rise correspondingly if the feature encodes a genuine clinical concept. We ground concordance within a broader validation framework that includes statistical grounding (point-biserial correlation against 46 ICD-9 codes with confound controls) and causal ablation (zero-ablation with differential loss measurement on diagnosis-positive versus diagnosis-negative notes). Together, these three modalities form converging evidence: grounding establishes that associations exist, concordance verifies that the explanations reflect them, and ablation confirms their causal relevance, supporting the conclusion that domain-trained SAE features encode real clinical concepts.

Our key findings are: (1) the concordance gradient rises from 85% to 98.6% as correlation strength increases, with features at $r > 0.5$ showing near-perfect alignment between their unsupervised explanations and the ICD code they track; (2) strongly grounded features are fully monospecific above $r > 0.6$, each encoding exactly one diagnosis code; (3) causal ablation confirms that grounded features differentially affect model predictions (median Cliff’s $\delta = 0.30$ for grounded features vs. -0.04 for controls); and (4) a general-domain SAE (GemmaScope) exhibits strongly negative reconstruction fidelity on clinical text ($EV = -4.21$), a result verified to stem from genuine feature-direction mismatch rather than mean-shift artifacts, indicating that domain-specific SAE training is needed for specialized domains.

We make three contributions:

1. **Concordance validation**, a general-purpose methodology for evaluating SAE features against external structured labels, applicable wherever paired free text and structured annotations exist, such as clinical diagnosis codes,

legal case categories, gene ontology terms, or similar.

2. **A converging-evidence validation framework** combining statistical grounding, semantic concordance, and causal ablation, demonstrated at clinical scale (50,000 notes, 46 codes, 18,432 features) with confound controls and surface-text baselines.
3. **Empirical evidence that domain-specific SAE training is necessary** for meaningful feature recovery in specialized text, supported by the demonstrated failure of a general-domain SAE on clinical data and the grounding of domain-trained features against external labels.

2 Related Works

2.1 Critiques of SAE Evaluation

Leask et al. (2025) demonstrate through SAE stitching and meta-SAEs that learned features are neither complete nor atomic, challenging the assumption that SAEs recover a canonical feature set. Koznikov et al. (2026) further develop this concern in a synthetic setting with known ground-truth features: SAEs recover only 9% of true features despite 71% explained variance, and random baselines (untrained feature directions) match trained SAEs on interpretability scoring, sparse probing, and causal editing.

A separate line of critique targets auto-interpretability as an evaluation method. Heap et al. (2025) show that SAEs trained on randomly initialized transformers produce auto-interpretability scores and reconstruction metrics broadly similar to those from trained models, implying that high scores alone do not guarantee recovery of computationally relevant features. McCann (2026) formalizes this further with *descriptive collision*: many distinct features admit the same natural-language explanation, and detection-style scoring is provably invariant to this, meaning inflated scores can arise without genuine feature discrimination.

Ma et al. (2026) extend the critique to feature-level claims, showing that 45–90% of purported reasoning features identified by contrastive activation methods can be activated by simple token injection, suggesting reliance on lexical cues rather than computational structure. Across these critiques, a shared characteristic is their reliance on *internal* evaluation: metrics derived from the model’s own

representations or scored by LLMs operating on the same activations, leaving open the question of whether validation against external structured labels can provide a stronger basis for assessing feature quality.

2.2 Domain-Specific SAE Applications

The closest work to ours is O’Neill et al. (2025), who train JumpReLU SAEs on Gemma-2 representations of 195,000 clinical QA examples and find that domain-confined SAEs explain up to 20% more variance and produce more interpretable features than broad-domain alternatives like GemmaScope. Their finding that domain-specific training captures representational structure missed by general-domain SAEs directly motivates our work; we build on this observation by introducing a structured-label-based evaluation framework for quantifying and validating the clinical concepts these features encode.

Sainsbury et al. (2026) train TopK SAEs on a clinical-specific autoregressive model using MIMIC-IV coded event sequences, demonstrating progressive abstraction across transformer depth and showing that SAE features can outperform dense representations for event prediction under full-sequence probing, with dense representations remaining competitive once evaluation controls for sequence leakage. SAEs have also been applied to EEG signal analysis (Lehn-Schiøler et al., 2026), automatic speech recognition (Pluth et al., 2026), and single-cell foundation models (Kendiukhov, 2026). This growing trend toward domain-specific SAE evaluation reflects recognition that general-purpose metrics are insufficient and that external ground truth, where available, provides stronger evidence of feature quality.

2.3 Structured Labels and Interpretability in Clinical NLP

Clinical NLP has a long history of leveraging ICD diagnosis codes (structured labels assigned by human medical coders based on the full clinical encounter, independent of any automated text processing) as supervision for representation learning, automated coding, and model evaluation (Mullenbach et al., 2018; Vu et al., 2020; Yuan et al., 2022). Within mechanistic interpretability, probing studies have shown that LLM representations encode rich linguistic and factual structure (Tenney et al., 2019; Hewitt and Manning, 2019), but no prior work connects SAE-level feature decomposition

to structured medical diagnosis labels or validates individual features against external text-level annotations.

Our auto-interpretability pipeline builds on Paulo et al. (2024), who introduced scalable fuzzing and detection scoring for SAE feature explanations. These metrics measure whether an explanation predicts activation patterns, a useful sanity check whose limitations are now well-documented (§2.1). Our concordance validation methodology, detailed in §3, differs in evaluating the *semantic content* of explanations against external labels rather than their predictive accuracy over activations.

3 Methodology

In our study, we first build the experimental apparatus (§3.1–§3.3): a corpus of clinical activations, two domain-specific SAEs trained on those activations, a general-domain pretrained SAE for comparison, and three baselines. We then apply three feature-validation procedures (§3.4–§3.6): statistical grounding against diagnosis labels, semantic concordance between unsupervised feature explanations and those labels, and causal ablation.

Part 1: Experimental Setup

3.1 Dataset and Activation Extraction

We use discharge summaries from MIMIC-IV-Note (Johnson et al., 2023a), the free-text note module of MIMIC-IV (Johnson et al., 2023b), a de-identified EHR database. Each summary is paired with the ICD-9 codes assigned to its admission, which serve as our external labels. Because MIMIC-IV mixes ICD-9 and ICD-10, we restrict to ICD-9-coded admissions for a single consistent label space. We draw 50,000 summaries from the training and validation splits, never the test split, enriched for uncommon diagnoses rather than matched to population prevalence: notes are grouped into four prevalence strata, and the sampling budget is weighted toward the rarer strata.

We extract residual-stream activations at layer 16 of Gemma-2-2B (Gemma Team et al., 2024). Layer 16 (~62% depth of this 26-layer model) sits in the mid-depth stream where abstract semantic features are expected (Tenney et al., 2019; Hewitt and Manning, 2019); and is one of the layers that GemmaScope provides, enabling a like-for-like comparison. The 50,000 notes yield about 15.2 million tokens at a maximum sequence length of 8,192, and we

mean-center the activations before training. A held-out set of 4,911 notes, about 10% of the corpus, is reserved for the ablation analysis, a grounding robustness check, and checkpoint selection for the vanilla SAE (§3.2). The appendix details the extraction precision, the centering procedure, and the automated data-quality checks.

3.2 SAE Models

A sparse autoencoder encodes an activation $x \in \mathbb{R}^{d_{\text{model}}}$ into a sparse code z and decodes a reconstruction $\hat{x}$, minimizing $\mathcal{L} = \|x - \hat{x}\|_2^2 + \lambda \mathcal{S}(z)$. We train two domain-specific SAEs on the centered activations, differing chiefly in their sparsity mechanism, and compare both against a general-domain SAE.

The **Vanilla ReLU SAE** (Bricken et al., 2023; Cunningham et al., 2023) computes the code $z = \text{ReLU}(\pi)$ from the pre-activation $\pi = W_{\text{enc}}(x - b_{\text{dec}}) + b_{\text{enc}}$, using an L_1 penalty for sparsity, and periodically resamples dead latents following Bricken et al. (2023). The **JumpReLU SAE** (Rajamanoharan et al., 2024) instead computes $z = \pi \odot H(\pi - \theta)$ from the same pre-activation, where H is the Heaviside step and $\theta \in \mathbb{R}_+^{d_{\text{sae}}}$ is a per-feature threshold; sparsity uses an L_0 penalty, and gradients pass through the threshold via straight-through estimators. Both SAEs use 18,432 latents, an eightfold expansion, constrain decoder columns to unit norm (Bricken et al., 2023), and optimize with Adam using $\beta_1 = 0$, following common SAE training practice. We select the best checkpoint by reconstruction explained variance, and the vanilla SAE uses the held-out set for selection and early stopping. The appendix tabulates the hyperparameters and training schedules (Table 5).

The **GemmaScope SAE** (Lieberum et al., 2024) is our general-domain baseline: the standard pre-trained SAE suite for Gemma-2. We use its layer-16 dictionary (width 16,384, nominal $L_0 \approx 42$) zero-shot on the clinical activations, without fine-tuning or re-centering. Because clinical text is out of distribution for this SAE, we additionally run domain-shift diagnostics (appendix) that distinguish a mean shift in the activations from a feature-direction mismatch.

For all three SAEs we report reconstruction fidelity on the held-out activations: explained variance, the mean number of active latents per token (L_0), and the dead-latent fraction.

3.3 Comparison Baselines

We compare the SAE features against three baselines, all sharing the SAE evaluation’s notes, pooling, and data split for a fair comparison.

The **lexical** baseline tests whether the features merely track keyword presence. For each code we build a per-note indicator of whether any curated keyword appears, drawn from an LLM-generated keyword dictionary, and compute its point-biserial correlation with the label. We compare this lexical correlation against the strongest SAE feature’s correlation for the same code, both before and after residualizing on note token count. We also measure keyword-absent recall: among label-positive notes containing no keyword, the rate at which the most associated feature still fires, testing whether the feature captures signal beyond lexical co-occurrence.

To ask whether a full-vocabulary classifier outperforms the features, the **TF-IDF** baseline fits per-code logistic regression on 10,000 unigram and bigram features with sublinear term frequency, under stratified 5-fold cross-validation, scored by AUC-ROC and AUC-PR. The same classifier is fit on the pooled SAE features; a paired Wilcoxon signed-rank test (Wilcoxon, 1945) over the per-code AUC differences compares the two, and a supplementary analysis contrasts the strongest single feature of each type.

The **raw-activation** baseline asks how much of the codes’ predictability is already linearly available in the dense activations, running the same cross-validated classifier on the max-pooled dense layer-16 activations as a reference point for the sparse-feature results.

Part 2: Validation Methodology

3.4 Statistical Grounding

We summarize each note by element-wise max-pooling its SAE activations across tokens, a standard step for downstream probing (Karvonen et al., 2025). Max-pooling couples feature magnitude with note length, a confound we control for below. We retain the 46 ICD-9 codes with prevalence above 2%. For each feature and code we compute the point-biserial correlation (Tate, 1954) between the pooled activation and the binary label (equivalently, the Pearson correlation with a dichotomous variable), with two-tailed p -values from the associated t -statistic. Because we test the full family of $d_{\text{sae}} \times 46$ associations, we apply the Benjamini–Hochberg procedure at $q = 0.05$ (Benjamini and

Hochberg, 1995). A feature is *grounded* if it has a significant association whose $|r_{pb}|$ exceeds a threshold, which we sweep from 0.1 to 0.7. Its specificity is the number of codes above threshold: monospecific (exactly one code), oligospecific (two or three codes), or polyspecific (four or more).

To control the length confound, we residualize each pooled activation on note token count by ordinary least squares, recompute the correlation, and re-apply the correction. The threshold sweep, the specificity counts, and the partial correlation are computed post hoc from the same correlations. We also recompute grounding on the held-out set as a robustness check. The appendix reports token-level inspection of grounded features and a case study of the top feature.

3.5 Semantic Concordance

Concordance asks whether two independently produced signals name the same clinical concept: an unsupervised explanation of a feature, and that feature’s statistical association with a diagnosis. We apply it to the JumpReLU SAE.

Explanations and scoring. Following the Delphi framework (Paulo et al., 2024), an explainer LLM, Claude Sonnet, writes a natural-language explanation of each feature from its top-activating token contexts together with non-activating contexts. Each context is a 30-token window on either side of the marked triggering token. The same model assigns each feature to one of five categories: clinical concept, clinical vocabulary, structural pattern, general language, or noise. We analyze 964 features in four tiers by grounding status: 280 strongly grounded ($|r_{pb}| > 0.4$), 100 weakly grounded ($0.1 < |r_{pb}| \leq 0.3$), 484 non-grounded, and 100 dead. We also compute Paulo et al.’s detection and fuzzing scores as a faithfulness check, but treat them as non-discriminative, since such scores do not separate trained from random transformers (Heap et al., 2025); a shuffled-explanation control (deranged explanations, chance ≈ 0.51) confirms the scorers respond to genuine explanation quality. Details are in the appendix..

Concordance validation. Consider a grounded feature with its strongest significant code c and its explanation e . A judge LLM rates whether e is semantically related to the natural-language description of c , taken from the ICD-9 reference, returning one of four verdicts: YES, the explanation clearly

describes the ICD concept; PARTIAL, the explanation overlaps but describes a related, broader, or narrower concept (e.g., a drug used to treat the condition, a symptom, or a broader category containing it); NO, the explanation describes something unrelated; or UNKNOWN, the judge cannot determine a relationship. The concordance rate is the fraction of YES and PARTIAL verdicts, reported at several $|r_{pb}|$ thresholds. The two signals are produced independently, since the explanation comes from activation contexts alone and the association comes from labels alone. This independence is what distinguishes concordance from scorer accuracy: an explanation can predict a feature’s activations without identifying a unique concept, a failure known as descriptive collision (McCann, 2026), whereas concordance tests whether the explanation’s semantic content matches an external label. Prompts are given in the appendix, and a single LLM is used throughout for explanation, scoring, and concordance.

3.6 Causal Ablation

To test causal relevance, we zero-ablate features: removing one that encodes a diagnosis should raise loss more on diagnosis-positive notes. At layer 16 we record the cross-entropy loss on a clean forward pass (ℓ_{clean}), after splicing in the SAE reconstruction (ℓ_{recon}), and after zeroing the target feature’s sparse code and decoding (ℓ_{abl}). Losses are measured over a window covering each note’s final 25% of non-padding tokens. The ablation effect of a feature is $\ell_{\text{abl}} - \ell_{\text{recon}}$, and the reconstruction tax is $\ell_{\text{recon}} - \ell_{\text{clean}}$. The targets are grounded features, selected by $|r_{pb}|$ and filtered for monospecificity and firing density, together with random and low- $|r_{pb}|$ control features; the appendix lists them. For each feature we compare ablation effects on label-positive against label-negative notes with a one-sided Mann–Whitney U test (Mann and Whitney, 1947). We report Cliff’s δ (Cliff, 1993) as the effect size, binned by magnitude following Romano et al. (2006), and apply Benjamini–Hochberg correction at $q = 0.05$.

4 Results

All correlation analyses use point-biserial r_{pb} with Benjamini–Hochberg FDR correction at $q=0.05$ (Benjamini and Hochberg, 1995). Unless stated otherwise, numbers refer to the JumpReLU SAE on the full 50,000-note corpus.

Table 1: Grounded latent counts by correlation threshold. Custom SAEs: 18,432 latents; GemmaScope: 16,384.

Threshold	JumpReLU	Vanilla	GemmaScope	Ratio
$ r > 0.1$	9,721	8,985	5,790	$1.6\times$
$ r > 0.3$	610	675	54	$12.5\times$
$ r > 0.5$	147	143	4	$36.8\times$
$ r > 0.7$	28	29	0	—
Peak $ r $	.864	.853	.574	

4.1 SAE Features Correlate with Diagnosis Codes

We compute the point-biserial correlation between each feature’s note-level activation (max-pooled across tokens) and each of 46 binary ICD-9 indicators. Table 1 and Figure 1 summarise the result.

Both domain-trained SAEs produce strong associations with specific diagnoses. The JumpReLU SAE achieves a peak $|r_{pb}|$ of 0.864 with atrial fibrillation (ICD-9 42731), Vanilla peaks at 0.853 with hypothyroidism (ICD-9 2449). GemmaScope-16k (Lieberum et al., 2024), applied zero-shot, peaks at 0.574. The gap widens at higher thresholds: at $|r| > 0.3$, domain-trained SAEs retain 610–673 grounded latents against GemmaScope’s 48 ($\sim 13\times$), and at $|r| > 0.5$ the ratio exceeds $35\times$. At $|r| > 0.6$ GemmaScope has no surviving features.

Both custom SAEs independently surface the same conditions at the top of the distribution (hypothyroidism, ESRD, atrial fibrillation), suggesting the grounding pattern reflects Gemma-2-2B’s representation of clinical text rather than SAE-specific artifacts. Full-corpus evaluation on all 50,000 notes produces consistent counts and peak correlations (Appendix C).

4.2 Concordance Between Feature Explanations and Structured Labels

We next test whether the grounding associations correspond to semantically coherent clinical concepts. For each grounded feature, we compare two independently generated signals: (i) an LLM explanation of the feature’s function, produced from top-activating token contexts with no access to ICD labels; and (ii) the ICD code with which the feature is most strongly correlated. A concordance judge (Claude Sonnet) evaluates whether the explanation describes a concept related to the code, assigning a verdict of YES (direct match), PARTIAL (semantically related), NO, or UNKNOWN. The concordance

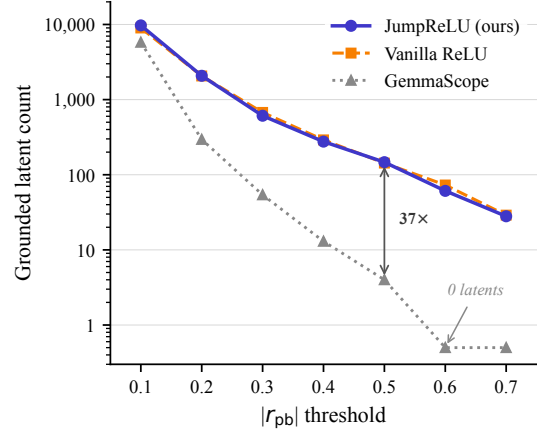


Figure 1: Grounded latent count vs. correlation threshold. GemmaScope reaches zero at $|r| > 0.6$.

rate is defined as $(\text{YES} + \text{PARTIAL})/N$, computed at multiple $|r|$ thresholds.

The concordance gradient. Table 2 and Figure 2 present the central result. At $|r| > 0.1$, concordance stands at 85.0%. At $|r| > 0.3$, it reaches 94.6%, with NO verdicts falling from 46 to 7. At $|r| > 0.5$, concordance is 98.6%: only 1 of 144 features receives a NO verdict, and the exact-match rate rises to 42.4%. This gradient shows us that two independently generated signals converge with increasing strength as the underlying clinical signal grows stronger.

Weakly grounded features ($|r|=0.1\text{--}0.3$) yield 58.0% concordance with zero exact matches and 39% NO, providing a built-in negative control: if concordance arose from generic “clinical-sounding” LLM explanations, it would not track correlation strength. This contrast also speaks to the descriptive collision concern raised by McCann (2026): if the LLM assigned the same boilerplate explanation to many features, the concordance rate would be flat across thresholds rather than rising from 85% to 99%.

Table 2: Concordance between LLM feature explanations and top-associated ICD-9 code (JumpReLU SAE).

$ r $ thresh.	N	YES	PARTIAL	NO	Concord.	Exact
> 0.1	380	85	238	46	85.0%	22.4%
> 0.3	280	85	180	7	94.6%	30.4%
> 0.5	144	61	81	1	98.6%	42.4%

Concord. = $(\text{YES} + \text{PARTIAL})/N$. UNKNOWN excluded.

Monospecificity. The concordance gradient is explained in part by a corresponding gradient in fea-

ture specificity. The fraction of grounded features tracking exactly one ICD code rises from 30.1% at $|r| > 0.1$ to 93.8% at $|r| > 0.5$ and 100% at $|r| > 0.6$ (mean codes per latent: 1.06 at $|r| > 0.5$). Features that encode a single diagnosis are those whose LLM explanations can unambiguously name that diagnosis.

Feature categorisation. An independent 5-way LLM categorisation of 964 features provides further confirmation. Of the 280 strongly grounded features ($|r| > 0.4$), 47.1% are categorised as *clinical_concept* (disease pathophysiology, treatment logic) and 42.9% as *clinical_vocabulary* (medical terms, abbreviations). Among the 484 non-grounded features, 99.8% are categorised as *noise*. Dead features (near-zero activation variance) are 75% *structural_pattern*—they fire on discharge note templates and section headers rather than random tokens.

Scorer accuracy. Standard fuzzing (0.938) and detection (0.962) scores are uniformly high across grounded, weakly grounded, and dead features, with no tier separation. This is consistent with [Heap et al. \(2025\)](#), who find that auto-interp scores do not discriminate between SAEs trained on functional versus randomly initialised transformers. We therefore treat these metrics as a sanity check on explanation faithfulness rather than a measure of feature quality, and rely on the concordance gradient (§4.2) for validation.

A shuffled-explanation control confirms the scores are still informative within that limited role. Each feature’s held-out contexts are re-scored against a wrong explanation drawn from a different feature, reproducing the random-baseline protocol of [Paulo et al. \(2024\)](#). Shuffled scores drop to chance (fuzzing: 0.496; detection: 0.522), while real scores hold at 0.932 and 0.961, a gap of ~ 0.44 (Wilcoxon $p < 10^{-6}$). The gap is stable across tiers, including dead features. The scorer is sensitive to explanation content; the uniformly high real scores reflect the specificity of the generated explanations rather than an inability to penalise poor ones.

4.3 Confound Controls and Domain-Shift Analysis

We test whether the observed correlations can be explained by note length, surface keywords, or a supervised text classifier.

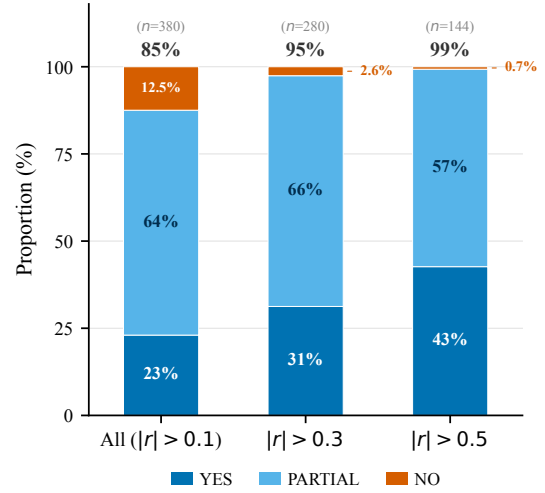


Figure 2: Concordance verdict distribution at increasing $|r|$ thresholds. NO verdicts shrink from 12% to under 1%.

Note-length confound. Max-pooling selects the element-wise maximum activation across tokens, so longer notes have more chances for high values. We control for this by computing partial correlations with note length (n_{tokens}) residualised via OLS. After this control, 41% of weakly grounded JumpReLU features ($|r| > 0.1$) lose significance ($9,721 \rightarrow 5,711$), but the top correlation barely moves ($0.864 \rightarrow 0.853$, a 1.3% drop). At $|r| > 0.5$ the picture reverses: 124 of 147 features survive (84%; Vanilla retains 92%), indicating that the strongest clinical associations are largely length-independent. GemmaScope loses 82% of its features at $|r| > 0.1$ ($5,790 \rightarrow 1,039$), and its mean max $|r|$ drops 46%. The strongest clinical features in domain-trained SAEs are length-independent.

Lexical baseline. Custom SAEs beat a curated keyword baseline (~ 25 keywords per code) on 45/46 codes (JumpReLU) and 44/46 (Vanilla), with mean $\Delta r = 0.287\text{--}0.308$. GemmaScope loses on 12/46. SAE features also fire on ICD-positive notes where no dictionary keyword appears, confirming that the SAE encodes information beyond lexical co-occurrence ([O’Neill et al., 2025](#)).

TF-IDF logistic regression. A TF-IDF logistic regression baseline (10,000 n -gram features, 5-fold CV) achieves higher classification AUC than SAE-based LR (mean AUC-ROC: 0.917 vs. 0.888). This gap is expected: TF-IDF trains a supervised classifier with a full vocabulary, while the SAE comparison wraps unsupervised features in a clas-

sification head.. A fairer single-feature comparison examining univariate feature quality shows the SAE’s best feature achieves a stronger correlation than TF-IDF’s on 21/46 codes (mean best- r : 0.566 vs. 0.519), consistent with the pattern reported by Sainsbury et al. (2026), who find that dense representations match or exceed SAE features for prediction tasks in clinical settings.

GemmaScope domain shift. GemmaScope’s explained variance on clinical activations is -4.21 , showing reconstruction is worse than predicting the global mean. Correcting for mean shift does not help: substituting the clinical mean for GemmaScope’s decoder bias leaves EV essentially unchanged (-4.20), and fully recentering input activations worsens it (-6.56). The cause is direction mismatch. Clinical activations project onto GemmaScope’s encoder directions predominantly with the wrong sign, producing large negative pre-activations that do not fire. Sparsity is nominally preserved ($L_0 \approx 50$), but the resulting sparse codes carry little clinical information, as the grounding and ablation results confirm. Our recentering analysis extends O’Neill et al. (2025), who report an EV advantage for domain-confined SAEs, by identifying the mechanism as direction mismatch rather than a correctable distributional shift.

4.4 Causal Ablation

To test whether grounded features are causally relevant rather than merely correlated we use zero-ablation: setting each feature’s sparse code to zero, decoding to a modified activation, splicing it into the residual stream at layer 16, and measuring the cross-entropy change on the last 25% of each note’s tokens. If a feature causally encodes a diagnosis, ablation should increase loss more on diagnosis-positive notes. We quantify this with Cliff’s δ (Cliff, 1993) and test with Mann-Whitney U (BH-FDR corrected).

The 10 Vanilla SAE features with the strongest ICD-9 correlations are all significant ($q < 0.05$; Table 3), with median $\delta = 0.300$ and three exceeding $\delta = 0.5$ (large effect, all hypothyroidism). Two control features (one random, one low- $|r|$) both produce null effects. An extended run of 30 features (top-10 plus ranks 11–30) yields 25/30 significant (83.3%), with median δ declining at lower ranks as expected. Three features (10%) show reverse-direction effects ($\delta < 0$), a reminder that correlational grounding does not guarantee a positive

Table 3: Causal ablation (4,911 held-out notes).

	Vanilla	GemmaScope
Sig. (top-10)	10/10	7/10
Med. δ (top-10)	.300	.195
Largest δ	.675	.562
Sig. (all 30)	25/30	22/30
Recon. tax (nats)	.029	.648
Controls	null	null

causal contribution.

The Vanilla SAE’s reconstruction tax is 0.029 nats (1.8% of base loss). GemmaScope yields 22/30 significant, but at a reconstruction tax of 0.648 nats ($22\times$ larger), its ablation measures removal from an already-degraded representation. Ablation for the JumpReLU SAE was not run; given the near-identical grounding profiles of the two architectures (Table 1), comparable results are expected but remain unconfirmed.

5 Conclusion

We introduced concordance validation, a methodology that tests whether unsupervised SAE feature explanations align with external structured labels the explanation pipeline never observed. Training sparse autoencoders on Gemma-2-2B representations of 50,000 clinical discharge summaries and validating against ICD-9 diagnosis codes, we found converging evidence across three modalities: statistical grounding (peak $r = 0.864$, full monospecificity above $r > 0.6$), a concordance gradient in which explanation–label alignment rises from 85% to 98.6% with increasing correlation strength, and causal ablation confirming differential loss on diagnosis-positive notes. A general-domain SAE fails on the same data because its feature directions do not span the clinical activation subspace.

The concordance gradient constitutes the central result. Because the feature explanation and the diagnosis label are produced by independent processes, their increasing convergence at stronger correlations provides evidence that the feature encodes the clinical concept the label names. This methodology extends to any domain with paired free text and structured annotations, and we encourage the community to apply it as a complement to existing SAE evaluation.

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A Data and Preprocessing

Data availability. MIMIC-IV discharge summaries are available through PhysioNet (<https://physionet.org>) under a credentialled data use agreement and cannot be redistributed.

Sampling. We drew 50,000 discharge summaries from the MIMIC-IV train and validation splits (test split never accessed); notes shorter than 500 characters were dropped. To ensure adequate representation of uncommon diagnoses, we stratified notes into four prevalence strata by the frequency of each note’s rarest active top-50 ICD-9 code (quartile bins), with per-stratum sampling budgets of 35/30/25/10 % (very-rare to common); deficit carried forward. Seed 42. Table 4 summarizes the corpus.

Table 4: Corpus statistics.

Quantity	Value
Notes	50,000
Tokens (total)	15,172,037
Shards	312
Max sequence length	8,192
ICD-9 codes (prevalence > 2%)	46
Held-out evaluation notes	4,911 ($\approx 10\%$)

Activation extraction. We extracted post-block residual-stream activations at layer 16 (`hidden_states[17]`) of Gemma-2-2B in fp32 and stored them as fp16 sharded safetensors. Tokenizer truncation was set at 8,192 tokens with special tokens. Post-extraction checks verified shape, finiteness, non-zero activation, and inter-note diversity.

Mean centering. A two-pass procedure computes the exact global mean: a float64 accumulator sums over all tokens, and the resulting mean is stored as float32. Centered shards are written as fp16. Centering is required for the domain-trained SAEs; without it, training produces a dominant DC-offset feature that captures the mean rather than meaningful variation.

B SAE Training Configuration

Both domain-trained SAEs share the architecture and optimizer settings in Table 5. They differ in their sparsity mechanism: the **Vanilla** SAE uses ReLU with an L_1 penalty ($\ell_1 = 10$) and resamples dead neurons at step 5,000 (firing threshold 10^{-6}); the **JumpReLU** SAE uses a learned-threshold

JumpReLU with an L_0 penalty ($\lambda_{L_0} = 10$), bandwidth $\varepsilon = 1.0$, initial log-threshold -2.0 , and a λ_{L_0} warmup of 5,000 steps to prevent premature threshold collapse. Both were trained for 3 epochs with the best checkpoint selected by explained variance. The Vanilla SAE early-stopped at 30,000 steps (patience = 3 evaluation rounds); the JumpReLU best checkpoint fell at step 36,000.

Table 5: Shared training parameters (domain-trained SAEs).

Parameter	Value
Input dimension d_{in}	2,304
Latent dimension d_{sac}	18,432 ($8\times$ expansion)
Learning rate	2×10^{-4}
Adam β	(0.0, 0.999)
LR warmup steps	2,000
Epochs	3
Batch size (tokens)	4,096
Decoder constraint	unit-norm
Checkpoint selection	test-set explained variance

The third comparator, **GemmaScope-16k** (Lieberum et al., 2024), is a pretrained general-domain JumpReLU SAE for Gemma-2-2B layer 16 with 16,384 latents and nominal $L_0 \approx 42$. We applied it zero-shot with no fine-tuning or recentering.

Table 6: Reconstruction fidelity (held-out activations).

	JumpReLU	Vanilla	GemmaScope
Explained var.	0.906	0.889	−4.21
Mean L_0	40.92	47.57	50.22
Dead-latent frac.	0.0262	0.0001	0.0276

Reconstruction fidelity. Table 6 reports reconstruction metrics for all three SAEs. GemmaScope’s strongly negative explained variance is analyzed in Appendix G.

Hyperparameter calibration. Before each production run, we measured the activation scale (σ) on centered activations and adjusted the sparsity penalty to target $L_0 \in [20, 80]$ on a 2,000-note calibration run before committing to the full corpus.

C Extended Grounding Results

C.1 Raw max-pool grounding and monospecificity. Table 7 reports, at each correlation threshold, the number of grounded latents and their monospecificity profile (mono = exactly 1 associated ICD code, oligo = 2–3, poly = 4+) for all

Table 7: Raw max-pool grounding with monospecificity profile (test split, 4,911 notes). Mono = 1 associated ICD code; Oligo = 2–3; Poly = ≥ 4 . Highest total per row in bold.

$ r $ thr.	JumpReLU		Vanilla		GemmaScope	
	Total	M/O/P	Total	M/O/P	Total	M/O/P
0.1	9,721	2,904/3,245/3,572	8,985	2,810/2,874/3,301	5,790	1,268/1,491/3,031
0.2	2,075	1,270/645/160	2,063	1,276/621/166	295	224/64/7
0.3	610	484/117/9	675	551/118/6	54	47/7/0
0.4	276	244/31/1	291	263/27/1	13	11/2/0
0.5	147	136/11/0	143	135/8/0	4	4/0/0
0.6	61	61/0/0	73	73/0/0	0	—
0.7	28	28/0/0	29	29/0/0	0	—

three SAEs using raw max-pooled activations. The highest total in each row is set in bold. The progression from 30% to 100% monospecific parallels the concordance gradient (§4.2): features that encode a single diagnosis are those whose LLM explanations can unambiguously name that diagnosis.

C.2 Partial correlation grounding and monospecificity. Table 8 repeats the analysis of Table 7 after residualising per-note SAE activations on n_{tokens} via OLS, removing the max-pooling length confound. The retention rate (partial count / raw count) at each threshold quantifies how much of the raw grounding signal survives the length correction.

C.3 Top-5 associations per SAE. Both domain-trained SAEs surface the same three conditions at the top of their association distribution (hypothyroidism 2449, end-stage renal disease 5856, atrial fibrillation 42731), suggesting the grounding pattern reflects Gemma-2-2B’s clinical representations rather than SAE-specific artifacts. GemmaScope’s top features spread across more codes but at markedly lower $|r|$ (peak 0.574 vs. 0.864).

D Auto-Interpretability and Concordance

D.1 Prompt templates. All LLM calls use Claude Sonnet (claude-sonnet-4-6). Below are the prompt templates used at each stage.

Explanation + Categorisation (20 top-activating contexts per feature, trigger token marked with asterisks):

```
Below are the top {n} text
contexts where a sparse
autoencoder feature activates
most strongly in clinical
discharge summaries. The
trigger token in each context
is marked with **asterisks**.

{contexts}

1. What pattern causes this
```

```
feature to activate? Describe
the specific concept, linguistic
pattern, or structural element
that these contexts share. Be
as specific as possible -- name
clinical concepts, drugs, or
procedures if applicable. One
to two sentences.
```

```
2. Categorize this feature into
exactly one of:
- clinical_concept (a specific
medical diagnosis, condition, or
finding)
- clinical_vocabulary (drug
names, dosages, units, lab
values, procedures)
- general_language (POS, syntax,
common words, generic patterns)
- structural_pattern (formatting,
section headers, signatures,
templates)
- noise (no coherent pattern)
```

```
Format your response as:
EXPLANATION: <your explanation>
CATEGORY: <category_name>
```

Fuzzing (token-level: can the LLM identify which tokens would activate the feature?):

```
A sparse autoencoder feature has
been described as:
"{explanation}"

For each of the following tokens
in context, predict whether this
token would activate the feature
(YES or NO).

{contexts}

Respond with one line per
context: the context number
and YES or NO.
```

Detection (example-level: can the LLM identify which text contexts activated the feature?):

```
A sparse autoencoder feature has
been described as:
"{explanation}"

Below are {n} text contexts from
clinical discharge summaries.
Some activated this feature and
```


Table 8: Partial-correlation grounding (n_{tokens} residualised) with monospecificity profile (test split, 4,911 notes). Highest total per row in bold.

$ r $ thr.	JumpReLU		Vanilla		GemmaScope	
	Total	M/O/P	Total	M/O/P	Total	M/O/P
0.1	5,711	2,674/1,811/1,226	5,355	2,557/1,685/1,113	1,039	639/267/133
0.2	1,276	887/328/61	1,335	941/322/72	125	101/21/3
0.3	451	357/89/5	503	416/83/4	39	35/4/0
0.4	236	208/27/1	257	237/19/1	12	11/1/0
0.5	124	117/7/0	131	126/5/0	3	3/0/0
0.6	54	54/0/0	64	64/0/0	0	—
0.7	27	27/0/0	27	27/0/0	0	—

some did not.
{contexts}
For each context, predict
whether it activated the feature
(YES or NO). Respond with one
line per context: the context
number and YES or NO.

Concordance judgement (checks alignment between the unsupervised explanation and the statistically associated ICD code):

A sparse autoencoder feature has
been auto-interpreted as:
"{explanation}"
This feature also has the
strongest statistical
correlation (point-biserial r
= {r_pb:.3f}) with the ICD-9
diagnosis code: {icd_code}
({icd_description}).
Does the auto-interp explanation
describe the same concept as the
ICD code? Respond with:
- YES -- the explanation clearly
describes the ICD concept
- PARTIAL -- the explanation
overlaps but describes a
related, broader, or narrower
concept (e.g., a drug used
to treat the condition, a
symptom of the condition, or
a broader category containing
the condition)
- NO -- the explanation
describes something unrelated
Format: <verdict> |
<one-sentence rationale>
Example: PARTIAL | The
explanation describes warfarin
dosing, which is a treatment for
the correlated ICD code (atrial
fibrillation), not the condition
itself.

D.2 Concordance by threshold. Table 9 gives the full concordance detail.

D.3 Five-way categorisation by tier. Table 10 confirms that grounding status corresponds to interpretable clinical content.

Table 9: Concordance verdicts by correlation threshold.

$ r $ thr.	Total	YES	PARTIAL	NO	Concordance
All (> 0.1)	380	85	238	46	85.0%
> 0.3	280	85	180	7	94.6%
> 0.5	144	61	81	1	98.6%

Table 10: Feature categorisation by grounding tier (%).

Category	Strong	Weak	Dead	Non-gr.
clinical_concept	47.1	14.0	4.0	0
clinical_vocabulary	42.9	39.0	20.0	0
structural_pattern	9.3	35.0	75.0	0.2
general_language	0.7	10.0	1.0	0
noise	0	0	0	99.8

D.4 Scorer summary and shuffled-explanation control. Table 11 reports fuzzing and detection scores by tier. Scores are uniformly high across all tiers, with no separation between strongly grounded and dead features, which is consistent with [Heap et al. \(2025\)](#). This motivates our use of concordance validation (which *does* discriminate: 85% $\rightarrow$ 99%) as the primary semantic validation modality.

Table 11: Fuzzing and detection scores by tier.

Tier	n_{fuzz}	Mean fuzz	n_{det}	Mean det
Global	432	0.938	459	0.962
Strong grounded	268	0.938	275	0.960
Weak grounded	94	0.941	90	0.955
Dead	69	0.932	93	0.975
Non-grounded	1	1.000	1	1.000

To verify that high scores are explanation-specific and not an artifact of the scoring task’s difficulty, we randomly permuted explanations across features and re-scored (Table 12). Consistent with prior results from [Paulo et al. \(2024\)](#), shuffled scores collapse to near-chance levels (~ 0.50), con-

Table 12: Shuffled-explanation control (fuzzing). Chance reference ≈ 0.51 (Paulo et al., 2024).

Tier	Mean real	Mean shuffled	Δ	95% CI (shuffled)	Wilcoxon p
Overall	0.932	0.496	0.436	[0.489, 0.502]	$< 10^{-6}$
Strong grounded	0.932	0.500	0.432	[0.493, 0.507]	$< 10^{-6}$
Weak grounded	0.935	0.489	0.445	[0.473, 0.506]	$< 10^{-6}$
Dead	0.923	0.490	0.434	[0.469, 0.510]	$< 10^{-6}$

firming that the scorer is sensitive to explanation content. The gap is stable across all tiers, including dead features.

E Causal Ablation

E.1 Ablation protocol. For each target feature j , we run three forward passes per note through Gemma-2-2B: (1) *clean* — unmodified model, yielding loss $\mathcal{L}_{\text{clean}}$; (2) *reconstruction* — at layer 16, the residual stream is replaced with the SAE reconstruction $\hat{\mathbf{x}} + \boldsymbol{\mu}$, yielding $\mathcal{L}_{\text{recon}}$; (3) *ablated* — feature j ’s sparse code is set to zero before decoding, yielding $\mathcal{L}_{\text{abl}}$. All losses are cross-entropy computed over the final 25% of each note’s tokens (the “loss window”), targeting the portion of discharge summaries where diagnostic content is concentrated. The *ablation effect* is $\Delta_j = \mathcal{L}_{\text{abl}} - \mathcal{L}_{\text{recon}}$, isolating feature j ’s contribution from the baseline reconstruction distortion. The *reconstruction tax* is $\mathcal{L}_{\text{recon}} - \mathcal{L}_{\text{clean}}$, averaged across all notes. Feature ablation is implemented as a subtraction from the decoded reconstruction: $\hat{\mathbf{x}}_{\text{abl}} = \hat{\mathbf{x}} - z_j \cdot \mathbf{W}_{\text{dec}}[j, :]$, which is mathematically equivalent to zeroing the latent before decoding but avoids redundant decoder work.

E.2 Statistical testing. For each feature j associated with ICD code c , the ablation effects Δ_j are partitioned into label-positive notes (the admission carries code c) and label-negative notes (it does not). A one-sided Mann–Whitney U test (alternative="greater") tests whether ablation effects are systematically larger on label-positive notes. p -values are corrected for multiple testing via Benjamini–Hochberg FDR at $q = 0.05$. Effect size is Cliff’s $\delta = (2U - n_+n_-)/(n_+n_-)$, binned as negligible ($|\delta| < 0.147$), small (< 0.33), medium (< 0.474), or large (≥ 0.474). Two control features per SAE (one random, one low- $|r|$) confirm that the methodology does not produce spurious positives; both controls correctly produced near-zero δ and failed to reach significance in both SAEs.

Table 13: Causal ablation headline comparison (top-10 grounded features).

Metric	Vanilla	GemmaScope
Grounded features significant (of 10)	10	7
Controls significant (of 2)	0	0
Median Cliff’s δ (grounded)	0.300	0.195
Reconstruction tax (nats/token)	0.029	0.648

E.3 Headline results. Table 13 summarizes the ablation results across both SAEs. Across the full top-10, the domain-trained vanilla SAE achieves a 100% causal-validation rate (10/10 significant at BH-FDR $q=0.05$), while GemmaScope validates 70% (7/10); controls are correctly null in both SAEs. The reconstruction tax confirms the domain-shift finding: vanilla’s reconstruction costs 0.029 nats/token to splice in, versus 0.648 for GemmaScope (a $22\times$ gap consistent with the EV difference of 0.89 vs. -4.2).

Tables 14 and 15 report ablation results for five representative features per SAE, selected from the top 10 by $|r_{pb}|$ for clinical diversity (avoiding redundant coverage of the same ICD code). All 4,911 held-out notes were used.

Table 14: Vanilla SAE: selected per-feature ablation results (top-10).

Feature	ICD code	r_{pb}	n_+	Cliff’s δ	p_{BH}	Mag.
3485	2449 (hypothyroidism)	0.831	565	0.672	2.8×10^{-221}	large
10520	5856 (renal failure)	0.860	208	0.428	4.3×10^{-26}	medium
10471	5856 (renal failure)	0.830	208	0.377	1.9×10^{-22}	medium
3569	42731 (atrial fib.)	0.831	794	0.096	4.7×10^{-7}	negl.
8823	2449 (hypothyroidism)	0.858	565	0.070	2.0×10^{-3}	negl.

E.4 Vanilla SAE: per-feature results. Table 14 reports five features selected from the top 10 to cover all three ICD codes represented (hypothyroidism, renal failure, atrial fibrillation) and to illustrate the range of effect sizes.

Feature 3485 (hypothyroidism, $\delta = 0.672$) shows the strongest causal effect in the experiment, while feature 8823, despite having the *highest* point-biserial correlation ($r = 0.858$), shows only negligible Cliff’s δ (0.070), demonstrating that correlation strength alone does not predict causal

involvement.

E.5 GemmaScope: per-feature results. GemmaScope’s top-10 covers nine distinct ICD codes; five are selected to span the full range from its strongest causal feature (1459, anticoagulant use, $\delta = 0.562$) to a correlational-only failure (8073, depression, $\delta = -0.058$, $p_{\text{BH}} = 0.998$). Feature 15439 (hypothyroidism, $\delta = 0.310$) enables a direct cross-SAE comparison with vanilla’s hypothyroidism features: vanilla’s best hypothyroidism δ is $2.2\times$ larger (0.672 vs. 0.310).

Table 15: GemmaScope: selected per-feature ablation results (top-10).

Feature	ICD code	r_{pb}	n_+	Cliff’s δ	p_{BH}	Mag.
1459	V5861 (anticoagulant)	0.538	483	0.562	8.2×10^{-99}	large
1100	V4986 (joint replace.)	0.431	305	0.451	1.1×10^{-39}	medium
12322	25000 (diabetes-2)	0.521	875	0.445	1.8×10^{-94}	medium
15439	2449 (hypothyroidism)	0.501	565	0.310	7.2×10^{-37}	negl.
8073	311 (depression)	0.495	674	-0.058	0.998	—

Feature 8073 exemplifies the correlational-only pattern: despite $r_{pb} = 0.495$, its mean activation on positive notes is only 0.027 (vs. 0.013 on negatives), and its mean ablation effect on positive notes is 8.7×10^{-5} nats/token, which is two orders of magnitude below the threshold at which causal effects become detectable ($\sim 10^{-4}$). Three of GemmaScope’s top-10 features (8073, 11693, 14478) share this pattern: reasonable correlation, weak absolute activation, and no detectable causal effect. This 30% correlational-only rate is the primary empirical argument for why intervention-based validation is necessary beyond correlational grounding.

F Baseline Details

F.1 Lexical keyword baseline. Keywords were generated by an LLM (64 codes attempted; the 46 overlapping with the prevalence-filtered evaluation set are used). Each code has ~ 25 keywords. Matching uses word-boundary regex for keywords ≤ 3 characters and case-insensitive substring otherwise. Length control: partial point-biserial residualised on token count. *Keyword-absent recall*: among label-positive notes containing no keyword, the fraction on which the best-associated SAE feature fires above its median activation.

E.2 TF-IDF + LR. Features: 10,000 (unigram + bigram, sublinear TF). Classifier: logistic regression (SAGA solver, L_2 regularisation, $\text{max_iter}=5,000$). Cross-validation: stratified

5-fold, seed 42. Metrics: AUC-ROC and AUC-PR per code, aggregated via mean and Wilcoxon signed-rank test across codes.

Table 16: TF-IDF vs. SAE LR: Wilcoxon signed-rank ($n=46$ codes).

SAE	Metric	Mean TF-IDF	Mean SAE	Med. Δ	p
Vanilla	AUC-ROC	0.917	0.881	-0.033	3.5×10^{-9}
Vanilla	AUC-PR	0.590	0.532	-0.059	6.2×10^{-8}
JumpReLU	AUC-ROC	0.917	0.888	-0.027	≈ 0
JumpReLU	AUC-PR	0.590	0.543	-0.048	5×10^{-6}

The full per-code TF-IDF vs. SAE comparison (46 codes) is available in the supplementary materials.

E.3 Raw-activation probe. Max-pooled centred layer-16 activations (2,304 dimensions); identical per-code stratified 5-fold LR protocol. Solo run (no SAE comparator): mean AUC-ROC = 0.808, mean AUC-PR = 0.353 across 46 codes. Head-to-head SAE-vs-raw comparisons are planned but not yet executed.

G GemmaScope Domain-Shift Diagnostics

Table 17 reports diagnostics for GemmaScope’s negative explained variance on clinical activations. We test whether the failure is due to a mean shift (the SAE’s decoder bias $\mathbf{b}_{\text{dec}}$ does not match the clinical activation mean) or to genuine feature-direction mismatch (the dictionary directions do not span the clinical activation subspace).

Table 17: GemmaScope domain-shift diagnostics.

Metric	Value
<i>Mean comparison</i>	
Clinical mean norm	131.704
$\mathbf{b}_{\text{dec}}$ norm	145.904
$\cos(\mathbf{b}_{\text{dec}}, \text{clinical mean})$	0.900
Relative difference	0.484
<i>Scale comparison</i>	
Frac. pre-activations above threshold	0.711
Clinical activation norm (mean $\pm$ std)	208.6 ± 63.2
Threshold mean $\pm$ std	6.86 ± 0.39
<i>Explained variance</i>	
EV — as published	-4.200
EV — clinical mean for decoder bias	-4.200
EV — fully recentred	-6.564

Replacing $\mathbf{b}_{\text{dec}}$ with the clinical mean does not change EV (both -4.20). Full recentring — subtracting the clinical mean before encoding and

adding it back after decoding — *worsens* EV to -6.56 . This rules out mean shift as the primary cause: the dictionary directions themselves do not align with the clinical activation subspace. The 71.1% of features firing above threshold (vs. $\sim 5\%$ for domain-trained SAEs) confirms that GemmaScope’s sparsity assumptions break down on out-of-distribution clinical text.

H Feature Inspection

Token-level feature inspection uses a two-pass algorithm: the first pass scans activation shards for the top- k tokens per grounded latent ($k = 50$, 20 shards sampled); the second pass re-tokenises matched notes to extract ± 5 -token context windows around each firing.

Table 18: Feature inspection summary.

Metric	Vanilla	JumpReLU
Latents inspected	20	20
Mean diversity score	0.091	0.087
Mean firing-rate ratio	10.39	13.62

Low diversity scores (0.087–0.091) indicate that top tokens per latent are lexically concentrated (e.g., variants of the same clinical term), consistent with monosemantic feature interpretation. JumpReLU’s higher mean firing-rate ratio (13.62 vs. 10.39) indicates sharper selectivity — its grounded latents fire disproportionately more on ICD-positive notes.

Table 19 reports per-latent details for the top-10 most strongly grounded features (JumpReLU). Token strings are shown without surrounding note context to avoid any risk of PHI exposure.

Table 19: Top-10 grounded latents: token-level evidence (JumpReLU). *To be populated from feature_inspection_report.json.*

Feature	ICD code	Top tokens	Fire ratio
<i>[10 rows — TBD]</i>			